DEPARTMENT OF DEFENSE

JOINT ORDNANCE TEST PROCEDURE (JOTP)-055

CRITERIA FOR SUBMUNITION ADVANCED FEATURES TO MEET THE 2017 DOD POLICY ON CLUSTER MUNITIONS

DOD Fuze Engineering Standardization Working Group (FESWG)

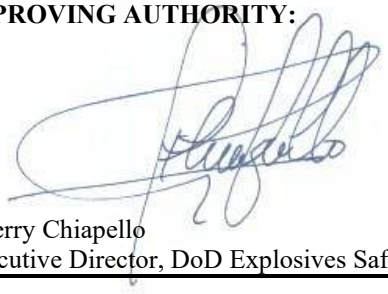
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Joint Ordnance Test Procedure (JOTP)-055
Criteria for Submunition Advanced Features to Meet
the 2017 DOD Policy on Cluster Munitions

DOCUMENT COMPLETION DATE: 26 January 2023	**TITLE AND SUBTITLE:** Joint Ordnance Test Procedure (JOTP)-055 Criteria for Submunition Advanced Features to Meet the 2017 DOD Policy on Cluster Munitions
PREPARING ACTIVITY: DOD Fuze Engineering Standardization Working Group U.S. Army DEVCOM Armaments Center ATTN: FCDD-ACE-Z / Building 6 Picatinny Arsenal, NJ 07806-5000	**SPONSORING ACTIVITY:** Policy and Standardization Division (CSTE-TM) US Army Test and Evaluation Command 6617 Aberdeen Boulevard Aberdeen Proving Ground, MD 21005-5001
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ABSTRACT: This document provides clarification and criteria for the implementation of fuzing system features used to meet the characteristics specified in paragraph 5.b in the Technical Specifications for the 2017 DoD Policy on Cluster Munitions.	
COORDINATION DRAFT REVIEWED BY: This document was coordinated with the following Standardization Offices: AR, AS, EA, MC, MI, MR, OS, TE, AF-2, AF-70, and AF-99. In addition, the document was also coordinated with the Joint Weapon Safety Working Group and select Subject Matter Experts (SMEs). **ASSIST COORDINATION DATE:** 09 September 2022	
IMPLEMENTATION PLAN: 1. This JOTP will be referenced in MIL-STD-1316, Fuze Design, Safety Criteria For. 2. This document has been developed by the DOD Fuze Engineering Standardization Working Group (FESWG) for use by the Army Fuze Safety Review Board (AFSRB), the Air Force Nonnuclear Munitions Safety Board (NNMSB), and the Navy Fuze and Initiation System Technical Review Panel (FISTRP) to evaluate and certify compliance with paragraph 5.b in the Technical Specifications for the 2017 DoD Policy on Cluster Munition.	
APPROVING AUTHORITY:  31 March 2023 Date Thierry Chiapello Executive Director, DoD Explosives Safety Board	

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JOINT ORDNANCE TEST PROCEDURE

Joint Ordnance Test Procedure (JOTP)-055 26 January 2023
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CRITERIA FOR SUBMUNITION ADVANCED FEATURES TO MEET THE 2017 DOD POLICY ON CLUSTER MUNITIONS

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1. SCOPE

1.1. This document provides clarification and criteria for the implementation of fuzing system features used to meet the characteristics specified in paragraph 5.b in the Technical Specifications for the 2017 DoD Policy on Cluster Munitions (CM). This document does not replace the fuze design safety criteria of MIL-STD-1316. Any fuze systems and/or ignition safety devices employed in the cluster munition shall conform to MIL-STD-1316 and/or MIL-STD-1901, as applicable.

1.2. This document has been developed by the DoD Fuze Engineering Standardization Working Group (DoD FESWG) for use by the Army Fuze Safety Review Board (AFSRB), the Air Force Nonnuclear Munitions Safety Board (NNMSB), and the Navy Fuze and Initiation System Technical Review Panel (FISTRP) to evaluate and certify compliance with paragraph 5.b in the Technical Specifications for the 2017 DoD Policy on CM.

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2. DEFINITIONS AND CLARIFICATION STATEMENTS

2.1. DEFINITIONS

For interpretation of this document, the following specific definitions apply.

2.1.1. Arm. In accordance with 2017 DoD Policy on CM and as specified herein, the term "Arm" is defined as a submunition warhead fuze system with the explosive train in an uninterrupted state and with sufficient accumulated energy capable of directly initiating the first explosive or pyrotechnic element in its fuze explosive train.

2.1.2. Function. The term "function", as defined in MIL-STD-1316, is equivalent to the term "detonate" used in the 2017 DoD Policy on CM.

2.1.3. Internal Power Source. A device or integral feature of the submunition capable of providing the energy required to initiate the fuze explosive train.

2.1.4. Submunition warhead. In accordance with 2017 DoD Policy on CM and as used within this JOTP, the assembly or subassembly containing the final or main explosive charge, inclusive of the fuze initiation system, which delivers the desired effects onto the intended target.

2.1.5. Submunition warhead subpack assembly. A payload assembly that houses submunition warheads and an integrated submunition warhead dispensing system used to dispense or eject the submunition warheads when intended.

2.1.6. Submunition. The submunition is either the submunition warhead or a submunition warhead subpack assembly dispensed or ejected directly from the delivery body or non-reusable canister.

2.1.7. Sterilization. A design feature that permanently prevents a fuze from functioning.

2.1.8. Submunition lifecycle environments. Any credible environment that the submunition may be exposed to during its lifecycle to include pre-launch/pre-deployment, launch/deployment and post-launch/post-deployment phases or events. This includes transportation, handling, storage, climate (temperature/humidity/pressure), electromagnetic fields, launch /deployment shock, flight dynamics/vibration, submunition expulsion, mid-air collisions, submunition warhead ejection, submunition warhead flight dynamics/vibration, impact and/or subsequent penetration (e.g., hard surfaces, soft terrains, foliage/trees, water submersion, etc.). The list is not all-inclusive and specific additional lifecycle environments must be considered.

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2.2. CLARIFICATION STATEMENTS

The following section provides clarification to paragraph 5.b of the 2017 DoD Policy on CM. Texts in italics are extracted directly from the 2017 DoD Policy on CM, while non-italicized texts are JOTP-055 statements.

2.2.1. 2017 DoD Policy on CM, paragraph 5.b.1 states:

"Each submunition or submunition warhead[(1)] is equipped with an internal power source that is essential for arming [(2.a, 2.b)] and detonation[(3)]. The submunition or submunition warhead is not designed to be detonated by mechanical means alone[(4)]."

The following clarification is provided:

a. [(1)] The term "submunition warhead" includes the fuze initiation system.

b. [(2.a)] For a submunition warhead fuze system employing a non-interrupted explosive train, the term "arming" is equivalent to the term "Armed" defined in MIL-STD-1316.

c. [(2.b)] For a submunition warhead fuze system employing explosive train interruption, the term "arming" is equivalent to explosive train interruption removed (reference paragraph 5.3.3 of MIL-STD-1316F) and the accumulation of the firing energy sufficient for the initiation of the first explosive or pyrotechnic element in the fuze submunition warhead explosive train.

d. [(3)] The term "detonation" is equivalent to the term "Function" defined in MIL-STD-1316.

e. [(4)] The term "by mechanical means alone", means a mechanical element(s), or electro-mechanical element(s) that responds to a force or inertial impulse on ground and/or target impact to directly initiate or generate energy to initiate the fuze explosive train.

2.2.2. 2017 DoD Policy on CM, paragraph 5.b.2 states:

"Each submunition or submunition warhead is equipped with at least one automatically functioning, electronic self-destruct mechanism that is in addition to the primary arming and detonation mechanism and that is designed to destroy the submunition or submunition warhead on which it is equipped, if the submunition or submunition warhead is not detonated by the primary arming and detonation mechanism."

The following clarification is provided:

One safety and arming device is sufficient for a submunition that provides multiple function modes that consist of a primary function mode (e.g., height of burst, point detonate, etc.) and, at a minimum, one secondary electrically-initiated self-destruct mechanism (e.g., time or point detonation) that is independent of the primary function mechanism.

2.2.3. 2017 DoD Policy on CM, paragraph 5.b.3 states:

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"Each submunition or submunition warhead that does not detonate or self-destruct is, once armed, rendered inoperable in 15 minutes or less by means of the irreversible exhaustion of a component (e.g. power source) that is essential to the operation of the submunition or submunition warhead."
No further clarification is required.

2.2.4. 2017 DoD Policy on CM, paragraph 5.b.4 states:
"Each submunition or submunition warhead that does not arm after being deployed from the non-reusable canister or delivery body cannot be subsequently armed or detonated by incidental handling, contact or movement."
No further clarification is required.

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3. REFERENCED AND RELATED DOCUMENTS

3.1. REFERENCED DOCUMENTS

a) 2017 DoD Cluster Munition Policy. *
b) MIL-STD-331, Fuzes, Ignition Safety Devices and Other Related Components, Environmental and Performance Tests For.
c) MIL-STD-461, Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment.
d) MIL-STD-1316, Fuze Design, Safety Criteria For.
e) DoD JOTP-052, Guideline for Qualification of Fuzes, Safe and Arm (S&A) Devices, and Ignition Safety Devices (ISD).
f) STANAG 4157, Safety, Arming and Functioning Systems (SAF Systems) Testing Requirements.
g) AOP-20, Safety, Arming and Functioning Systems Manual of Tests.

3.2. RELATED DOCUMENTS

a) MIL-STD-464, Electromagnetic Environmental Effects Requirements for Systems.
b) MIL-STD-1901, Munition Rocket and Missile Motor Ignition System Design, Safety Criteria For.
c) DoD JOTP-051, Technical Manual for the use of Logic Devices in Safety Features.
d) DoD JOTP-053, Electrical Stress Test.
e) DoD JOTP-062, Personnel-borne ElectroStatic Discharge (PESD) and Helicopter-borne ElectroStatic Discharge (HESD) Requirements for Ordnance.
f) DoD JOTP-072, Requirements for Development of Explosive Ordnance Disposal Procedures.
g) MIL-HDBK-1455, Dispenser and Sub-Munition, Air-Delivered, Safety Design and Safety Qualification Criteria For.
h) AOP-4157, Safety, Arming and Functioning Systems (SAF Systems) Testing Requirements.

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4. REQUIREMENTS

Any submunition that is properly dispensed from the cluster munition and whose energetic materials are not fully expended/consumed as intended, shall be considered unexploded ordnance (UXO) that must conform to the 2017 DoD Policy on CM. The following requirements establish the design and evaluation criteria for the advanced features cited in paragraph 5.b in the Technical Specifications for the 2017 DoD Policy on CM.

4.1. Internal Power Source

4.1.1. Upon arming or intended functioning, internal power source(s), and any resulting stored energy required to initiate the fuze explosive train, shall be permanently depleted or the submunition warhead shall be rendered inoperable by other means within 15 minutes.

4.1.2. Internal power source(s) shall not be capable of arming the submunition warhead fuze, if it did not previously arm, or functioning the explosive train due to the effects of UXO handling environments.

4.2. Target Sensor

4.2.1. Target sensors that employ mechanical or electromechanical devices (e.g., inertial or spring-driven fire pins, piezo-electrical devices, etc.) to directly initiate the fuze explosive train shall be prohibited.

4.3. Submunition warheads subpack assembly

4.3.1. Submunition warhead(s) that fail to eject as intended from a submunition warhead subpack assembly shall not arm or function if manually removed from its subpack assembly and/or by subsequent incidental UXO handling, contact or movement.

4.3.2. An integrated dispensing system (i.e., ignition system) used within the submunition warhead subpack assembly that fails to eject submunition warhead(s) as intended shall not subsequently eject the submunitions due to incidental UXO handling, contact or movement, if ejection results in either the submunition warhead(s) arming and/or functioning or other unacceptable hazards.

4.4. Electronic self-destruct mechanism

4.4.1. Self-destruct functionality shall remain operable after exposure to all submunition lifecycle environments.

4.4.2. Self-destruct mode shall function within 15 minutes of arming or shall be permanently inoperable after 15 minutes in accordance with paragraph 5.b.3 in the Technical Specifications for the 2017 DoD Policy on CM.

4.5. Irreversible exhaustion of a component essential to operation

4.5.1. The feature(s) used to meet paragraph 5.b.3 in the Technical Specifications for the 2017 DoD Policy on CM shall remain operable after being subjected to all submunition lifecycle environments. Examples of allowable feature(s) that render the submunition inoperable through irreversible exhaustion of a component include, but not limited to:

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a. Self-exhausted power source within 15 minutes.
b. A permanent load drain circuit across a power supply within 15 minutes.
c. Sterilization of the first explosive or pyrotechnic element in the fuze explosive train.
d. Fusible link within the submunition to interrupt power.
e. Permanent shorting of fire capacitor(s).
f. Continuous dissipation of fire capacitor(s) energy below the no-fire level.
g. Permanently interrupting the explosive train to prevent submunition warhead(s) functioning or other unacceptable hazards.

4.6. Electrical firing energy dissipation

4.6.1. Submunition advanced features shall not negate or compromise the MIL-STD-1316F, paragraph 4.6.4, requirement for electrical firing energy dissipation.

4.7. Submunition advanced features failure rates

4.7.1. The submunition advanced features failure rate shall be calculated for a submunition warhead fuze system that has been exposed to all submunition life cycle environments. The failure rate shall be verified to the extent practical by test and analysis and shall not exceed the rates given for the following:

a. For a submunition warhead fuze system that arms and fails to function by the primary arming and detonation mechanism, the total failure rate to either self-destruct (2017 DoD Policy on CM, paragraph 5.b.2) or be rendered inoperable (2017 DoD Policy on CM, paragraph 5.b.3) shall be less than one in a thousand.
b. For any submunition warhead fuze system that fails to arm, the failure rate to subsequently prevent arming or functioning due to incidental UXO handling, contact or movement (2017 DoD Policy on CM, paragraph 5.b.4) shall be less than one in a thousand.

4.8. Analyses

4.8.1. System analyses shall be conducted to arrive at an estimate of the system failure rates and to identify any credible failure modes of the submunition advanced features when exposed to all submunition lifecycle environments. These analyses shall be used to ensure the following:

a. Design features used to meet the 2017 DoD Policy on CM requirements of 5.b.2 and 5.b.3 remain operable.
b. Submunition that does not arm cannot be subsequently armed or functioned (i.e., detonated) per 2017 DoD Policy on CM requirements of 5.b.4.

4.8.2. Detailed analyses such as fault trees as well as failure modes, effects and criticality analysis shall be conducted as a minimum to identify failure modes for the purpose of their elimination or control.

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4.9. Design for quality control, inspection, and maintenance of submunition advanced features

4.9.1. Submunition advanced features shall be designed and documented to facilitate application of effective quality control and inspection procedures.

4.9.2. Design characteristics critical to meeting the 2017 DoD Policy on CM requirements of 5.b shall be identified.

4.9.3. If the approved design critical characteristics change, confer with the appropriate Service Safety Authority (SSA) to obtain concurrence that the intended design functionality has been maintained.

4.10. Documentation

The evaluation program prepared by the developing agency for use as the basis of compliance to meet the 2017 DoD Policy on CM requirements of 5.b shall be documented in both detail and summary form.

4.10.1. Critical components and characteristics

Components of the submunition advanced features with characteristics that may be critical for compliance with the 2017 DoD Policy on CM requirements of 5.b shall be identified and appraised in a document that forms part of the submunition warhead fuzing system specification. The evaluation of the critical components shall be presented as part of the design compliance assessment. The applicable technical documents (drawings, specifications, purchase descriptions, etc.) shall be annotated to show that the component is critical and cite the associated characteristics.

4.11. Review Authority

4.11.1. At the inception of engineering development, the developing agency should obtain approval of the submunition warhead fuze system design and test and evaluation program from the appropriate SSA tasked with assuring compliance in meeting the 2017 DoD Policy on CM requirements of 5.b. Adequate design documentation, detailed analysis, and test plans shall be provided to the SSA for review and approval.

4.11.2. At the completion of engineering development, the developing agency shall present a compliance assessment to the appropriate SSA for review to obtain approval of the design. Adequate design documentation, detailed analyses, test reports and identification of critical characteristics shall be provided to the SSA for review and approval.

4.12. Qualification of submunition advanced features

Compliance with the 2017 DoD Policy on CM shall be demonstrated through tests and/or appropriate analyses. Testing shall include integrated dispensing systems if the system presents an unacceptable hazard per paragraph 4.3.2. Qualification test plans shall be submitted to the appropriate SSA for approval of the appropriate test selections and quantities to be tested to address the submunition lifecycle environments. Refer to

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Appendix A for an example of a notional test series used to evaluate submunition advanced features.

4.12.1. Sequential Environment Testing Methods in accordance with JOTP-052 shall be used as a guide for appropriate test selection and quantities to be tested to address the lifecycle environments associated with pre-launch/pre-deployment and launch/deployment. Testing shall be conducted in accordance with MIL-STD-331 and STANAG 4157 with AOP-20 as appropriate.

4.12.2. Test assets that were subjected to the requirements of 4.12.1 shall be subjected to additional tests that simulate submunition lifecycle environments associated with post-launch/post-deployment. At a minimum, the tests shall include flight vibration, dispense event, and impact, as applicable.

4.12.2.1. After completion of the testing specified in paragraphs 4.12.1 and 4.12.2, functional tests shall be conducted to demonstrate that the following remain operable:

a. Electronic self-destruct capability (reference paragraph 4.4).

b. Feature(s) that render the submunition fuze system permanently inoperable within 15 minutes (reference paragraph 4.5).

4.12.3. Test assets subjected to the requirements of 4.12.2 shall then be subjected to additional tests that simulate post-impact environments associated with incidental UXO handling, contact or movement. At a minimum, tests shall include 3-meter drop, transportation vibration (i.e., loose cargo, military wheeled vehicle), thermal cycling, Personnel Electrostatic Discharge (PESD), and radiated susceptibility (e.g., emissions from hand-held radios, cellphones, AM/FM radios, TV signals, etc.), as applicable.

a. For submunition warhead fuze system(s) that armed and subsequently rendered inoperable, testing shall demonstrate that the submunition warhead fuze system remains inoperable as part of the test plan after the lifecycle environments of 4.12.3.

b. Testing shall include submunitions that are not armed to demonstrate that the submunition warhead fuze system does not subsequently arm or function. For a submunition warhead fuze system(s) employing explosive train interruption, the test samples shall be configured with the explosive train interruption removed. Secondary explosive elements may be omitted or replaced by inert mass simulants as agreed by the SSA.

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Appendix A

Qualification Test Method

Figure 1 provides an example of a notional test series used to evaluate submunition advanced features. It is expected that detailed tests and test items shall be based on unique program specifications and submunition lifecycle environments. To meet the entire MIL-STD-1316 submunition fuze qualification requirements, separate test units may be required to address the test criteria for both fuze safety qualification (JOTP-052) and submunition advanced features (JOTP-055). The final test plan shall be presented to the SSA for review and approval.

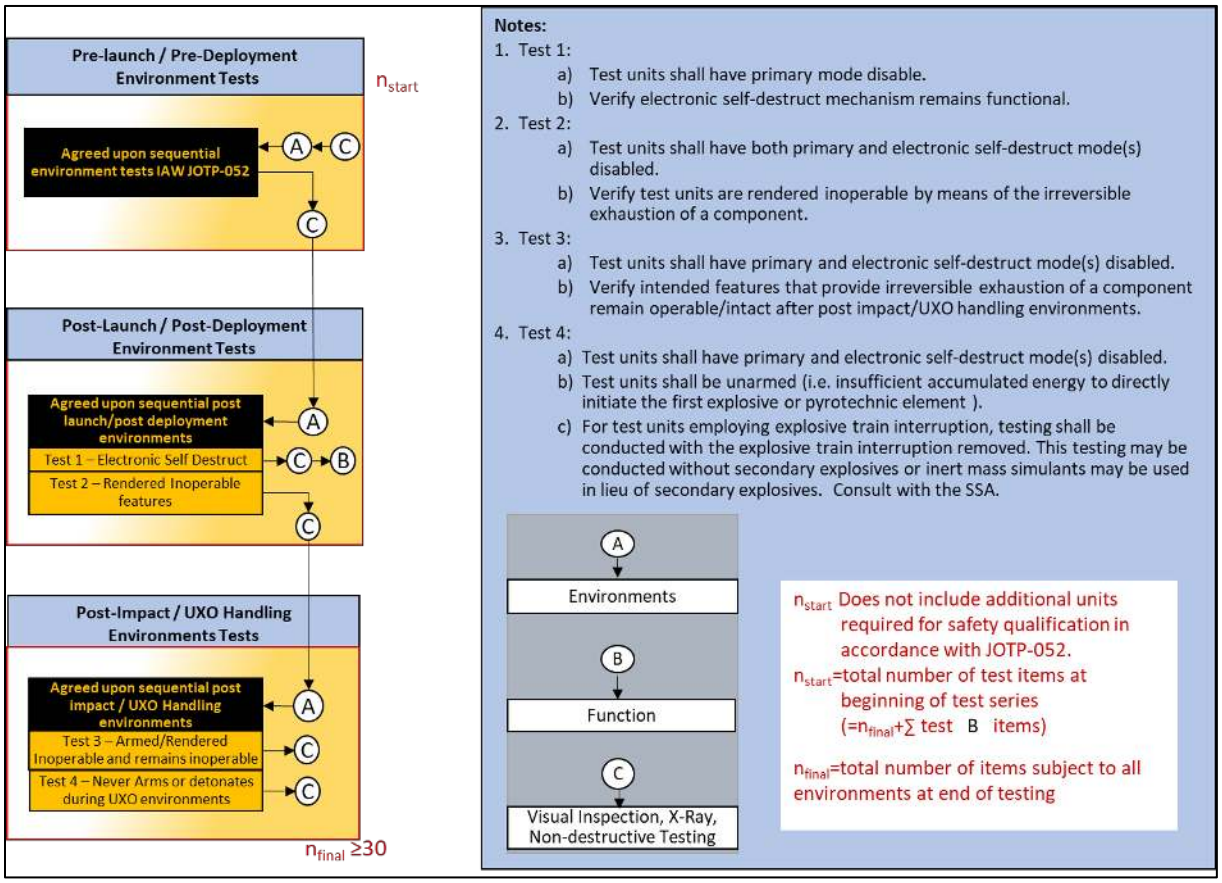


Figure 1 - Test series

The following list provides examples of submunition lifecycle environments associated with post-launch/post-deployment.

a. Flight Vibration
b. Pre-dispense: Pyrotechnic shock events
c. Dispense Event
d. Post Dispense Events:

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i. Air Stream dynamics
ii. Drag Force
iii. Submunition Collision (self-destruct features and/or “advanced features” must survive (pass criteria))
iv. Impact shock of submissions colliding
v. Survive Sympathetic destruct of features due to proximity of detonation
vi. Stabilizer (e.g., air deployed ribbon, parachute, etc.) or decelerator failure
vii. Terrain Impact
viii. Water Impact

The following list provides examples of submunition lifecycle environments associated with post-impact; incidental UXO handling, contact or movement:

a. Water submersion.
b. Temperature and humidity in accordance with test C1 of MIL-STD-331 exposed to a single 14-day duration.
c. 3-meter drop using the procedures of Test A4.1 of MIL-STD-331.
d. Jumble in accordance with test A2.2 of MIL-STD-331.
e. PESD and recommended radiated susceptibility tests in accordance with MIL-STD-461 RS103 except use a continuous wave signal source (no modulation).

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